IRAS Research Data Management Guidelines

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Preface

This book serves as a guideline for research data management at the Institute for Risk Assessment Sciences (IRAS) of the Faculty of Veterinary Science. It outlines and defines a set of principles regarding research data and provides practical handlers for researchers in accordance with the regulations established in the University Policy Framework for Research Data from Utrecht University (UU) and the Research Data Management Guidelines developed by the Faculty of Veterinary Medicine. These guidelines can be used as a reference together with existing UU policies, faculty guidelines and/or procedures already established by research groups.

> **! Important**
>
> Please note that the UU's University Policy Framework for Research Data dates back to 1 January 2016. This is before e.g. the GDPR was established. Hence, some of the information mentioned in the policy framework is not up to date. UU is in the process of creating a new policy framework. In the meantime, please consult RDM Support in case of doubt about the information in the current policy framework.

Part I

Introduction

The current chapter describes:

- Chapter 1: how this book is structured.
- Chapter 2: why good research data management is important.
- Chapter 3: responsibilities of everyone involved in the research project.
- Chapter 4: where you can find support.

1 Structure of this book

The IRAS Research Data Management guidelines are structured according to the Research Data Life Cycle (Figure 1.1). This cycle can be broadly divided into three phases: before (i.e. planning and designing), during and after research. Research data management (RDM) activities and actions take place during all three phases.

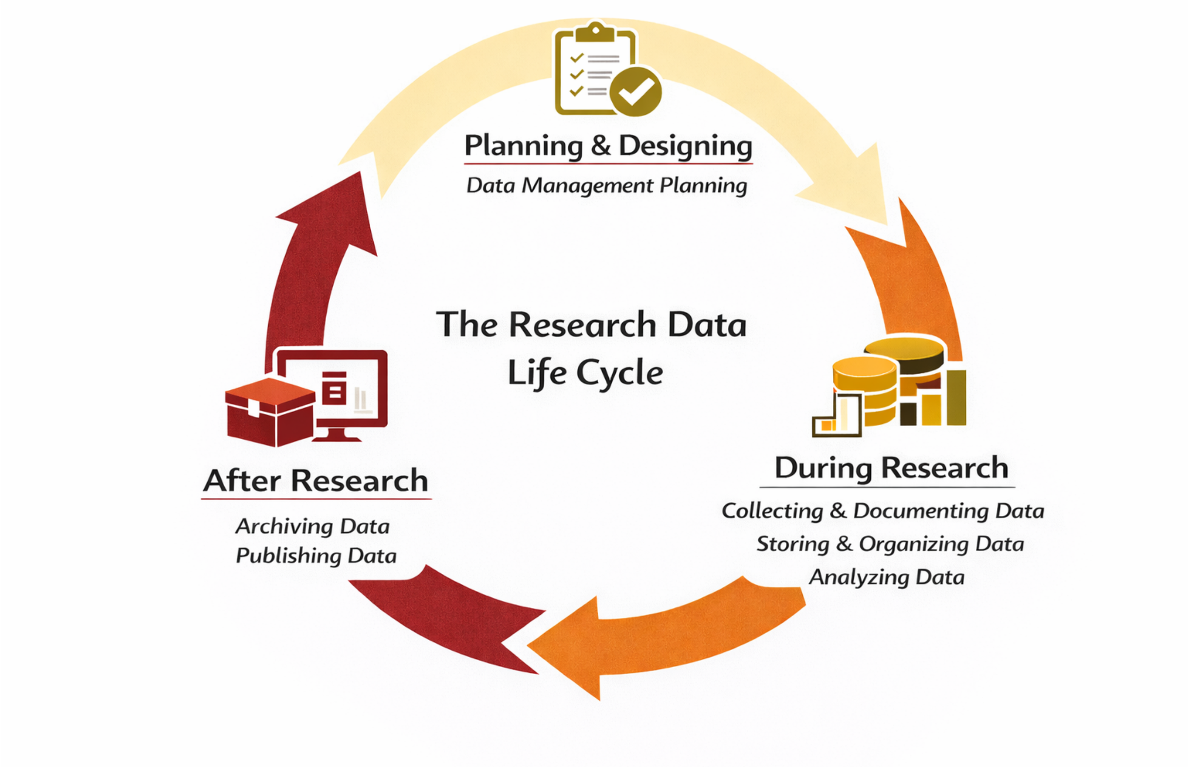

Figure 1.1: The Research Data Life Cycle

2 Importance of good research data management (RDM)

RDM refers to how you handle, organize, and structure your research data throughout the research process. Good RDM is essential for ensuring the integrity and validity of research, facilitating reproducibility and promoting collaboration. It also helps protect sensitive information and comply with regulations. Key reasons why good RDM is important include:

- **Ensuring data quality and security:** proper RDM prevents data loss, protects against unauthorized access, and ensures that research results are reliable and valid.
- **Increasing efficiency:** organizing, documenting, and structuring data saves time and resources by making it easier to find, understand and use data. Not only for others, but also for your future self.
- **Enabling reproducibility and validation:** well-managed, documented data allows other researchers to verify findings, contributing to the overall reliability and validity of scientific knowledge.
- **Allowing for reuse and collaboration:** making data understandable for others facilitates collaboration by enabling researchers to easily share and access data, accelerating scientific discovery.
- **Compliance and integrity:** good RDM ensures compliance with legal (e.g. GDPR when working with personal data), ethical, and funder requirements, and adheres to scientific integrity standards.
- **Improving visibility and impact:** making data FAIR (Findable, Accessible, Interoperable, Reusable) increases citation potential and the overall impact of research.

Proper RDM covers the entire data life cycle, from initial planning and collection to storage, analysis, archiving, and sharing. It is essential for effective Open Science and FAIR practices.

2.1 Open Science

In 2017, in order to accelerate and improve the realization of research results and their societal impact, the UU decided to make the transition to Open Science. Open Science is about increasing the reuse of research, and making sure that publicly funded research is accessible to all. Key to achieving this is adhering to the FAIR principles: ensuring the findings and data behind research results are findable, accessible, interoperable, and reusable (Bruce and Cordewener (2018)).

2.2 FAIR principles

In line with the UU's commitment to Open Science, research data at IRAS is expected to follow the FAIR principles:

- **F - Findable:** data and metadata should be findable, preferably digitally.
- **A - Accessible:** data and metadata should be stored and maintained in such a manner that they are accessible and with clear terms of use.
- **I - Interoperable:** data and metadata data should be stored and maintained in such a manner that the data can be easily opened, exchanged and combined with other datasets.
- **R - Reusable:** data are licensed and described in a manner that facilitates its reuse by other users.

Guidance on how to make your research data FAIR is highlighted in the recommendations provided below. To learn more see the RDM Support guide on how to make your data FAIR.

3 Responsibilities

The **project leader** (e.g. principal investigator) is responsible for …

Depending on the project requirements, **project members** (e.g. researchers, PhD/MSc/BSc students, data managers, project managers, research assistants) should:

- sign a confidentiality agreement in case the project requires working with personal data;
- write a data management plan (DMP);
- …

4 Support

4.1 IRAS Data Management team

The IRAS Data Management team is available for consultation on matters related to data management. We have a wide variety of expertise within the team, from data collection and cleaning to working with GIS data and setting up and hosting applications. While capacity within the team is limited when it comes to providing hands-on support, we will most of the time be able to provide advice on any data related questions you might have or bring you in to contact with the right people to help you further. You can contact us at iras_dm@uu.nl.

4.2 UU RDM Support team

In addition to the data management team at IRAS, the UU's Research Data Management (RDM) Support team is there to support, advise and assist all researchers at UU with data management, data analysis, research software and code. A multidisciplinary network of data experts, software specialists and research engineers offer their expertise through training, tools, infrastructure, consultancy, and custom-made solutions at every stage of a research project. The support they provide complies with the FAIR principles: findable, accessible, interoperable and reusable. That way, it contributes to Open Science.

> **RDM Support workshops and guides**
>
> To learn more about how to manage your research data and software, the UU's RDM Support team offers several trainings and workshops, as well as walk-in events. For a complete event agenda, together with information on trainings and workshop, please refer to the RDM Support website.
> Reading guides on diverse RDM topics can also be found on the RDM Support website. The same goes for tools that may be useful.

4.3 Other support services

Besides the IRAS Data Management team and RDM Support team, the following support services exist at UU:

- Research Support Office (RSO) of the Faculty of Veterinary Medicine: each faculty has an RSO. The RSO supports researchers throughout the year in the entire process of acquiring grants. For example: advice, information, reading, examples, budget assistance, data management, ethical questions, administrative data, etc.
- Privacy Officers of the Faculty of Veterinary Medicine: if you are starting a new project or research involving the use of personal data, exchanging data with a third party, or have questions about the secure storage of data or GDPR, please contact the Privacy Officers. You could also contact them for questions about the safe and responsible use of AI.

4.4 Contact

In case you have questions about research data management, please reach out!

> **Contact details**
>
> IRAS Data Management team: iras_dm@uu.nl
> RDM Support team: info.rdm@uu.nl
> RSO of the Faculty of Veterinary Medicine: rso-dgk@uu.nl
> Privacy Officers of the Faculty of Veterinary Medicine: privacy-dgk@uu.nl

Part II

Planning and designing

The current chapter describes:

- Chapter 5: why you should write a DMP and how to do that.
- Chapter 6: why you should estimate the costs that will be associated with data management and how to do that.

5 Data management plan (DMP)

RDM Support has instructions for developing a DMP. At this page several questions about DMPs are answered, such as the ones below.

Questions and answers

- **What is a DMP?** A DMP is a formal document you develop at the start of your research project which outlines all aspects of managing your data, both during and after your project.
- **Why do you need a DMP?** Data management is all about making your work efficient and creating more value for your data for yourself and others, during and after your research, for instance by working towards FAIR (findable, accessible, interoperable, reusable) data.
- **How to write a data management paragraph for a grant proposal?** When writing a research grant proposal, funders, such as NWO, will often ask for a data management paragraph.
- **How to write a DMP?** When writing your own DMP, you can:
  - Use DMPonline to write your DMP: DMPonline contains templates from many different funders (e.g., NWO, ZonMw and ERC) as well as the institutional templates from participating universities. The template offered by Utrecht University is tailored to the requirements which are laid down in the University Policy Framework for Research Data And approved by both NWO and ZonMw.
  - Get advice or feedback: if you are in need of advice or want your DMP reviewed, you can contact Research Data Management Support anytime, or click "Request feedback" in DMPonline.
  - Need more information or an introduction? Follow the online training Learn to write your DMP or join the workshop Quick start to Research Data Management and ask all your questions.

The use of DMPonline is further explained at the following page: https://www.uu.nl/en/research/research-data-management/tools-services/tool-to-create-your-dmp-online.

5.1 Our recommendation

Our recommendation is to write a DMP at the start of your research project and use the applicable template in DMPonline. This contains several examples you can use for your project. If you have any questions or want to have your DMP reviewed, please contact RDM Support as outlined above. During onboarding this will be explained to new staff.

6 Costs

Research data management will likely incur financial costs. In particular, the costs of data storage, archiving and publishing can be high. In some cases, these costs may not be covered by the university, and thus they will need to be obtained through funding or via other routes during the planning phase of the project. Luckily, most funders consider the costs for data management eligible for funding.

> Support
>
> RDM Support has developed a **guide to estimate the costs of data management** for each research phase and research activity. The costs of research data management will be a lot lower when you plan well ahead, instead of applying it retrospectively, both in terms of time and money.
> Check out the RSO intranet page for help with the **funding and grant acquisition process**.

6.1 Our recommendation

Our recommendation is to use RMD Support's guide to estimate the costs involved in research data management before you start your research project (e.g. when you start writing your DMP or applying for funding), because this will save you time and money in the end. If you need support or assistance, please contact RDM Support.

Part III

During research

The current chapter describes:

- Chapter 7: what tools the UU offers to collect research data, how to use them and what to pay attention to in terms of data management.
- Chapter 8: how best to store and organize your research data during your research project.
- Chapter 9: why documenting your data is important and how to do this.
- Chapter 10: where best to analyze your data and how to use GitHub for version control.

7 Collecting data

7.1 Formdesk

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7.2 Informed consent

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7.3 Qualtrics

If you want to create a survey that will collect personal data, the UU has one option and that is Qualtrics. Qualtrics is free of charge and can be used to set up surveys, collect data, distribute and analyze responses. Issues such as privacy and security are well taken care of. For further information please refer to the Qualtrics intranet page of the UU. It tells more about privacy, rules of conduct, non-standard UU modules in Qualtrics and more.

You can log into the Qualtrics website with your Solis-id (or UU email address), Solis-password and two-factor authentication. Once you have logged into Qualtrics, you can create a new survey via the Projects page. The Project page shows all surveys you have created (or that have been shared with you). You can collaborate on surveys with others by sharing them.

Qualtrics is known for having its fair share of quirks. A lack of awareness of these can lead to frustrations when using it to collect data. Therefore, Appendix B offers recommendations and tips and tricks for building a Qualtrics survey in a FAIR and structured way, while keeping data quality and privacy in mind. This may take some investment upfront, but will pay off in the end.

8 Storing and organizing data

8.1 Storing data

Good data storage within a project is very important, and with the growth of the PHS-IRAS division it becomes even more important. The aim of these guidelines is to make storage more uniform across the different projects.

The reasons why good data storage is important are threefold:

1. With regard to identifying personal data: Privacy regulations (GDPR) and medical ethical clearance (METC) procedures make it necessary that data have to be carefully handled.
2. With regard to coded (pseudonymized) or anonymous data:
    - It is important that results can be reproduced.
    - It is important that errors will be prevented.
    - If there is a follow-up after the current project, it is necessary that the data have been clearly managed and stored.
    - In case of an audit the correct data should be easily obtainable.
3. With regard to both coded (pseudonymized) or anonymous data and identifying personal data:
    - When an employee leaves IRAS the data should be accessible, and should be managed and stored in an understandable way for others.
    - It is important that no data is lost.
    - The storage location must be secured against unauthorized access.

8.1.1 Tools and information

Within the UU several sources of information are available where to store your data safely. It depends upon the research phase and the type of data what an appropriate storage platform is. You can follow the UU checklist on storage to find the correct platform for your data. Below you will also find some concrete recommendations that apply to most research projects within IRAS.

https://www.uu.nl/en/research/research-data-management/guides/storing-and-preserving-data

https://tools.uu.nl/storagefinder/

Never use -the C/personal drive of your laptop or PC or other external hard drives (including USB sticks and SD cards): these can be lost or damaged, posing significant risks. Dropbox, Google Drive, or the consumer version of OneDrive: your data may be stored outside of Europe (and therefore not covered by European law), and the products are offered "for free" (so the companies monetize your content).

8.1.2 Data classification

To determine the necessary measures to protect your data, it is helpful to first perform a data classification of the data you will collect in your project. This will help you find a suitable data storage tool:

The goal of classifying information is roughly categorizing the potential impact for the university if the availability, integrity or confidentiality of the data get compromised. By answering a set of questions you will determine the value of your data and the security risk that your data is exposed to. You will conclude what the impact is that a breach can have on the availability (what if data cannot be accessed or the data is lost entirely), the integrity (what if the data is incorrect) and confidentiality (what if an unauthorized person gains access to the data and what if the data is leaked). Making this assessment is called data classification.

https://www.uu.nl/en/research/research-data-management/guides/legal-instruments-and-agreements

https://intranet.uu.nl/en/security/information-security-data-classification

Important considerations for storing personal data at IRAS

A short summary of the IRAS Data guidelines for working with personal data:

1. Personal data should only be accessible to those who need to work with it. This means that data (that is used only infrequently) should be stored centrally (where it can be properly secured) and only accessible to a limited number of people.

2. Personal data should not be stored on a local drive. Apart from unauthorized access, the risk of data loss is much higher with a local drive due to it crashing, getting lost, or being stolen than with a network drive (which is automatically backed up).

3. Personal data may only be accessed on secure UU devices. For example, it is not permitted to use personally identifiable information on personal computers, USB sticks, hard drives, or in the cloud.

It is very important that data (in general) does not end up in the hands of unauthorized persons. Data protection starts at the workplace, where simple measures can greatly reduce the chance that unauthorized persons obtain personal data. Therefore you should always:

1. Keep personal data, like informed consent forms from questionnaires, field forms with names and addresses, etc., under lock and key and separate from the coded (pseudonymized) or anonymous data.
2. It is important that access to PCs on which identifying personal data is processed is adequately protected. Processing of identifying personal data is only allowed on secure UU (portable) computers. Access to computers should be secured with a 2FA login-procedure with password, with an automatic log-off after a period of inactivity or a guarded screen-saver. Terminal equipment, such as printers, should also not be accessible to unauthorized persons.

Particular attention deserves employees that work at home. IRAS does only allow accessing or working with identifying personal data at home via VPN. Identifying personal data should not be transported by means of portable mediums. In exceptional cases, for single transport from institute to institute, for instance, together with samples, encrypted information should be carried on a password secured and labeled data stick.

8.1.3 IRAS storage recommendations

If you collect sensitive data within your project we recommend to use the UU network O-drive to store and analyze your data OR to use YODA to store your data and use a different analysis platform to analyze your data. All available storage locations with a high security that can be used for sensitive data within the UU are described in Excel file 'Storage Overview High Security.xlsx'.

8.1.4 O-drive

O-drive: For collaboration with colleagues and research groups within the faculty. Here you can store and share information with your colleagues. It's secure storage provided in a secure folder for which you've set permissions and which you then continue to monitor. Automatic backup of your data is provided in the form of snapshots that go back 90 days. The O-drive is not suitable for storing large data files, as it's very expensive for the faculty.

On the O-drive it is recommended to create a 'Privacy folder' (see section 2.5 of the IRAS guidelines) and a 'Project' folder for your project. Identifying data is stored in the project's 'Privacy' folder, with restricted access only to those who need this data. The pseudonymized research data can be stored in the 'Project' folder, with access for researchers performing the analysis. It is very important that you properly manage access rights to the folders on the O-drive. If necessary, you can create separate folders with restricted access, for e.g. Master's students. Or add a password to a logbook used by multiple people. The final data is archived in YODA. Folder structure is important. In the IRAS data guidelines a template is mentioned for the O-drive in section 4. Proposed structure of electronic project folder; 4.1. General structure.

8.1.5 YODA

YODA: is suitable for long-term collaboration and long-term storage of research data. Automatic backup is also provided. The Faculty of Veterinary Medicine promotes this storage for large amounts of research data. YODA is a research data management service that enables you to store, share, archive and publish your research data throughout the research lifecycle. Working with YODA is in line with the FAIR principles and open science. YODA is suitable for a wide variety of data types and formats, including sensitive personal data. You can explore published datasets using DataCite Commons: YODA Public Data. YODA enables you to collaborate with other researchers, both inside and outside your own institution. The software is developed at Utrecht University and is used by multiple organisations, both in the Netherlands and abroad. All information how to use YODA is available here: https://www.uu.nl/en/research/YODA

8.1.6 Proposed folder and file naming for YODA

Practical guidelines Yoda (i.e. technical limitations): - Research project folder names: o Only use lowercase letters, digits or hyphens - Filenames within folder: o case-sensitive and may also contain e.g. spaces (but never single quotes) - Limits on whole logical file path: max 1023 characters always including prefix '/nluu6p/home/research-' but user-usable limit is lower due to how Yoda works: at least 900 characters available (e.g. Yoda internally uses some extra postfixing of names/timestamped extensions, for revisions, the vault, and for files in the "trash" area) NB: limits can differ between operating systems when trying to access Yoda via e.g. WebDAV https://github.com/UtrechtUniversity/yoda/discussions/168 - Yoda does not (yet!) have explicit limits or usability guidelines for research project folder names

Data manager guidelines (i.e. preferred): - Research project folder names: o Preferably as short as possible o Includes project name   Other considerations (that not everyone uses) o Project name alone is hard to make meaningful (especially a short name that should last over time); so project folder names often also include the last name and initials of the researcher (alternative: solisID to make it unique) o A set project folder name structure allows for more overview when mounting Yoda as a drive (e.g. sorted order in Windows file explorer) o Community and subcommunity names are also used as a way to provide more information about the research project folders (e.g. project name in the community name and project folder names only using researcher names) o Because of the way access rights work in Yoda, often the same research project folder name is repeated with a suffix (e.g. adding '-shared') to share subsets of the data with different people - It would be very useful for continuity and easy transfer between data managers if every community contains a README file for data managers, describing the structure (i.e. community and subcommunity levels) and research project folder naming guidelines used in that community.

8.1.7 A proposed folder structure for YODA

• Create a main folder • With research groups, names starting with research-: (work folder for temporary data storage during the fieldwork period or analysis) (eventually for the data in the project folder at the O-drive) (eventually for the Privacy data at the O-drive) (to exchange data with external partners, access can be temporarily) • Subfolders can be made under the research groups, e.g.: projectname_dataset • projectname_raw_data • projectname_processed_data projectname_script projectname_document projectname_paper (his folder is created to publish data. Use a serial number if multiple papers have been published) • paper.pdf • dataset • script • codebook Readme.txt (This contains e.g. how the folder structure is structured)

The YODA team has started working on YODA v2.1.0 and have planned working on creating a folder structure from a template.

8.1.8 Access to the data in YODA

First you need to request access to Yoda: https://www.uu.nl/en/research/yoda/guide-to-yoda/requesting-access-to-yoda For detailed instruction for accessing your data in YODA: https://www.uu.nl/en/research/YODA/guide-to-YODA/i-want-to-start-using-YODA You can access the data in YODA in two ways: via a network drive in your file manager and through the website portal. If you want to download or upload bigger amounts of data or files (more than a couple of files / MB's), you need to use the YODA network drive instead of the website portal. • Within the network drive, you work with your files just as you normally would do while using your file explorer. You will have direct access to your files via your file explorer. To configure YODA as a network drive see configure the YODA network drive. For the network drive use: https://dgk.data.uu.nl
• In the web portal you can arrange who gets access to your data, archive data and publish data. This is not possible via a network drive. In addition, you can upload and download files, although this will be slower than using YODA as a network drive.
For the website portal go to https://dgk.YODA.uu.nl • You can also use iBridges to access

YODA: https://www.uu.nl/en/research/yoda/guide-to-yoda/accessing-yoda/using-ibridges-to-access-yoda YODA allows you to store your data in a secure way. All data in YODA is stored in research groups. These are the main folders. Their names always start with 'research-'. Now that you have access to YODA through the network drive, you can start working with your data in your research group. You can work with your data like you normally would via the file explorer. For example, you can copy, paste, drag, open, and edit your data. Use your existing applications and tools to work with the data. Contact person: Wouter Kok (use Topdesk external link to request access to YODA)

8.1.9 Analysis platforms

YODA is a storage solution but not really useful for data analysis. The UU had made an overview of which tools you can use from a security and privacy aspect. You can use the following tools to analyze your data: https://intranet.uu.nl/en/knowledgebase/which-tools-and-software-are-secure https://tools.uu.nl/tooladvisor/

8.1.10 Source data, adapted data, analysis data and versioning

Please see the IRAS Data Guidelines: 4.5.3 Availability of data and findings Raw data, without any manipulations, should always be stored as well as cleaned data. Datasets that are under construction can be stored in a "working" directory. At the end of the project, there should not be any files in the "working" directory and the "working" directory can be deleted. Different versions of a dataset should be clearly identifiable by adding a version-number or the date to the file name. It's important to periodically store the raw (source) data in a secure location to prevent loss. This can be done, for example, in YODA's Vault. Programs (e.g. SAS-, R- or SPSS-codes) for data cleaning/definition of derived variables should be stored and serve as documentation of the data cleaning process and definition of derived variables. Alternatively, the data cleaning and definition of derived variables can be documented in a Word-file. Different versions should be clearly identifiable by adding a version-number or the date to the file name. The same system for identification of different versions (version number/date) could be used for programs/Word documents and datasets, to have a clear link between the dataset and the corresponding program/document. 4.6 Analysis (for paper) folder All dataset(s), programs (e.g. SAS-, SPSS-, S-Plus, R-codes etc.), documents and figures related to a specific analysis (e.g. for a paper) should be stored together in a separate folder. That way results presented in a paper can be reproduced and future analysis on exactly the same data that has been used for an earlier paper can be done. Different versions of a dataset should be clearly identifiable by adding a version-number or the date to the file name. As long as the paper is work in progress, this folder should be stored on the used storage location. Once the paper has been accepted for publication, the folder should be moved to the project's final folder on the O-drive or in the Vault in YODA. Only final versions of datasets and programs should be stored here. In principle, this will be done by the project leader/data manager. To ensure that no changes will be made to the files stored on the O-drive, the analysis folder and its content will be stored as read-only or in the Vault in YODA where it is not possible to make adaptations.

8.1.11 Archiving and publishing of data

At the end of the project or when a paper has been summitted, you can archive and/or publish your data in YODA: https://www.uu.nl/en/research/YODA/guide-to-YODA/archiving-your-data-in-YODA https://www.uu.nl/en/research/YODA/guide-to-YODA/publishing-your-data-in-YODA

8.1.12 Metadata

If you have data, you should inevitably have metadata. Metadata and documentation provide information about your data, such as the topic of your project or the circumstances under which your data were obtained. Without metadata and documentation, a lot of data are just numbers without any meaning. Therefore, this “data about data” is essential to understand, find, reuse and manage (the context of) your data. https://www.uu.nl/en/research/research-data-management/guides/during-research/metadata-and-documentation

8.1.13 Data breach

There is a data breach in case when an incident involves the university’s employees or students or IT resources. If you suspect a data breach, please report it as soon as possible via databreach@uu.nl. https://intranet.uu.nl/en/knowledgebase/what-do-i-have-to-do-if-there-is-a-personal-data-breach https://www.uu.nl/en/organisation/practical-matters/privacy/report-data-breach

8.1.14 Links with additional information

For further information please see the following links: https://www.uu.nl/en/research/research-data-management/tools-services/tools-for-storing-and-managing-data https://intranet.uu.nl/en/information-security-where-should-i-store-my-data https://intranet.uu.nl/kennisbank/onderzoeksinformatie-dgk (Dutch) At Teams: Storage Overview High Security.xlsx

8.2 Organizing data

8.2.1 Folder structures

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8.2.2 File naming conventions

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8.2.3 Data formats

There is a lot of diversity in data formats. When possible, it is important to choose for future-proof data formats (i.e. open, standardized formats) to be able to continue reading, displaying and processing research data in the future.

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9 Documenting data

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10 Analyzing data

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Part IV

After research

The current chapter describes:

- Chapter 11: how best to archive your research data after your research project.
- Chapter 12: how best to publish your research data after your research project.

11 Archiving data

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12 Publishing data

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Part V

Data security and privacy

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13 Legal instruments and agreements, policies and laws

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14 Handling personal data

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15 Data security

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16 ...

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References

Bruce, R., and B. Cordewener. 2018. "Open Science Is All Very Well but How Do You Make It FAIR in Practice?" *LSE Impact of Social Sciences Blog.* London School of Economics & Political Science. https://blogs.lse.ac.uk/impactofsocialsciences/2018/07/26/open-science-is-all-very-well-but-how-do-you-make-it-fair-in-practice/.

Appendix A

List of mentioned tools

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Appendix B

Qualtrics tips & tricks

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